

Machine Learning

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Outline

- Neural Networks
- Perceptrons

Neural Networks

- A Neural Network (NN) is a network of neurons that are used to process information.
- A neural network is a group of interconnected units called neurons that send signals to one another.
- Neurons can be either **biological cells** or **mathematical models**.
- The most advanced data processing machine at the time — the brain.

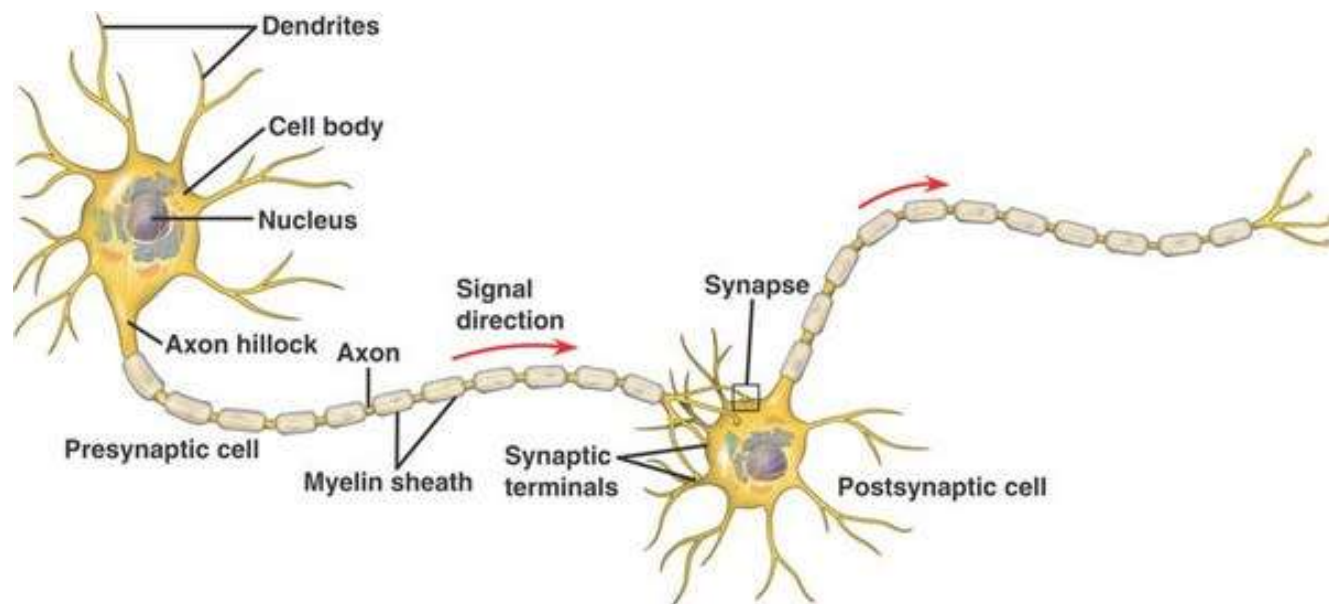
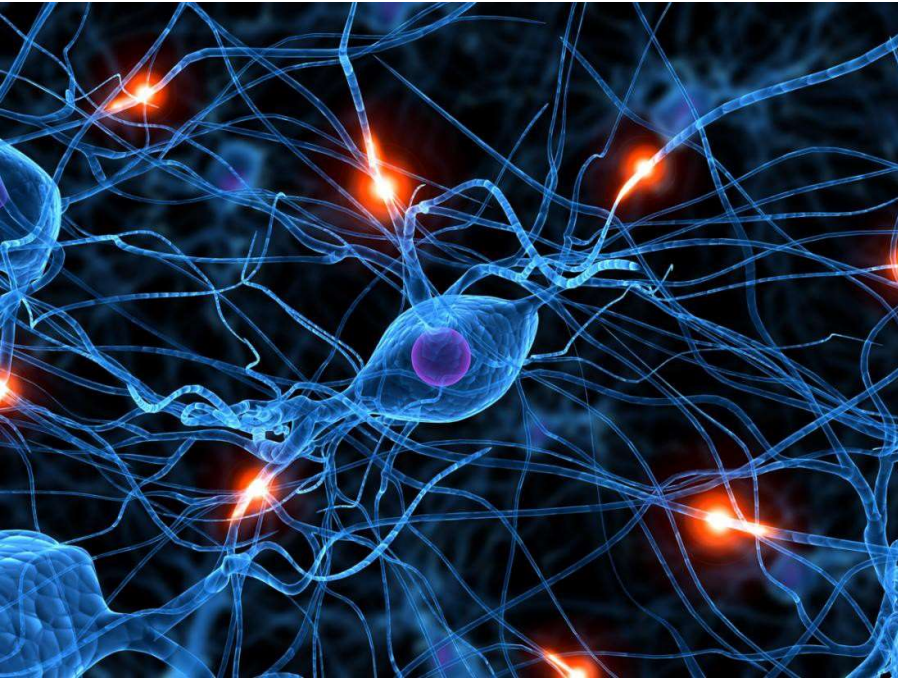
Types of Neural Networks

- In neuroscience, a **Biological Neural Network (BNN)** is a physical structure found in brains and complex nervous systems – a population of nerve cells connected by synapses.
- In machine learning, an **Artificial Neural Network (ANN)** is a **mathematical model** used to approximate nonlinear functions. Artificial neural networks are used to solve artificial intelligence problems.

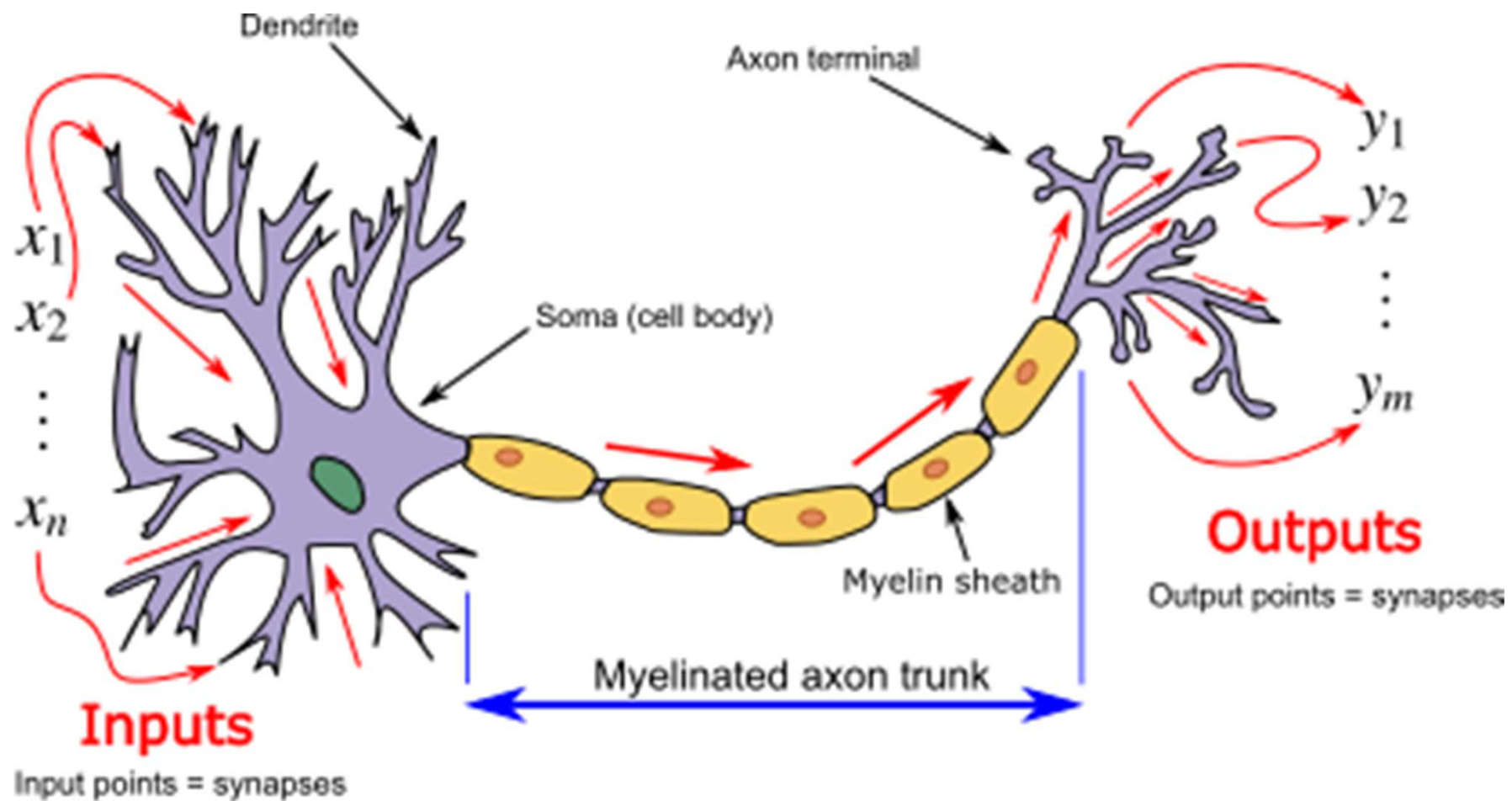
Biological Neural Networks

- In the context of biology, a neural network is a population of biological neurons chemically connected to each other by synapses.
- A given neuron can be connected to hundreds of thousands of synapses.
- Each neuron sends and receives electrochemical signals called action potentials to its connected neighbors.
- A neuron can serve an excitatory role, amplifying and propagating signals it receives, or an inhibitory role, suppressing signals instead.

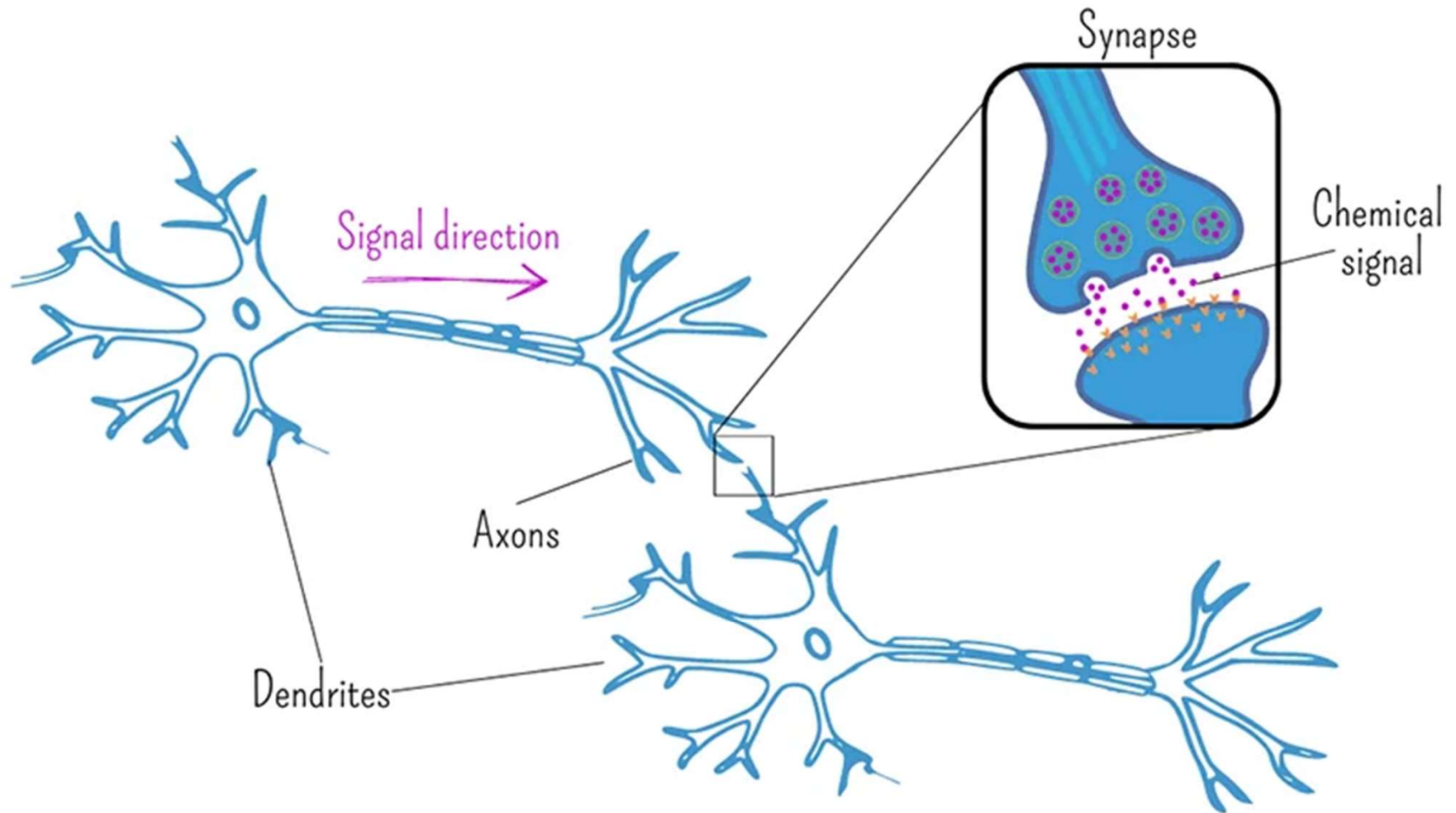
Biological Neural Networks



Biological Neural Networks



Biological Neural Networks



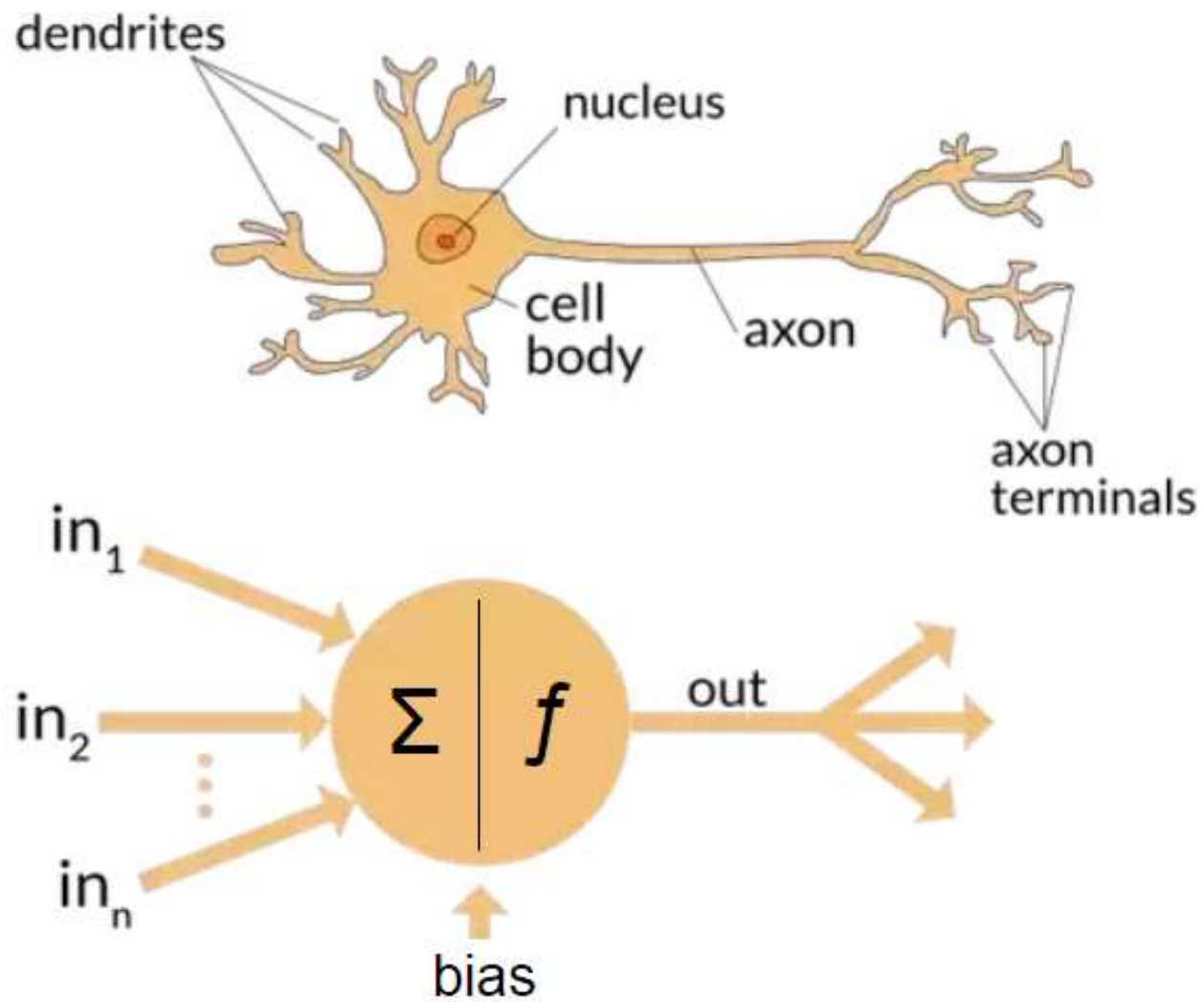
Biological Neural Networks

- The brain is principally composed of about 10 billion neurons, each connected to about 10,000 other neurons.
- Each neuron receives electrochemical inputs from other neurons at the dendrites.
- If the sum of these electrical inputs is sufficiently powerful to activate the neuron, it transmits an electrochemical signal along the axon and passes this signal to the other neurons whose dendrites are attached at any of the axon terminals. These attached neurons may then fire.
- It is important to note that a neuron fires only if the total signal received at the cell body exceeds a certain level. The neuron either fires or it doesn't, there aren't different grades of firing.

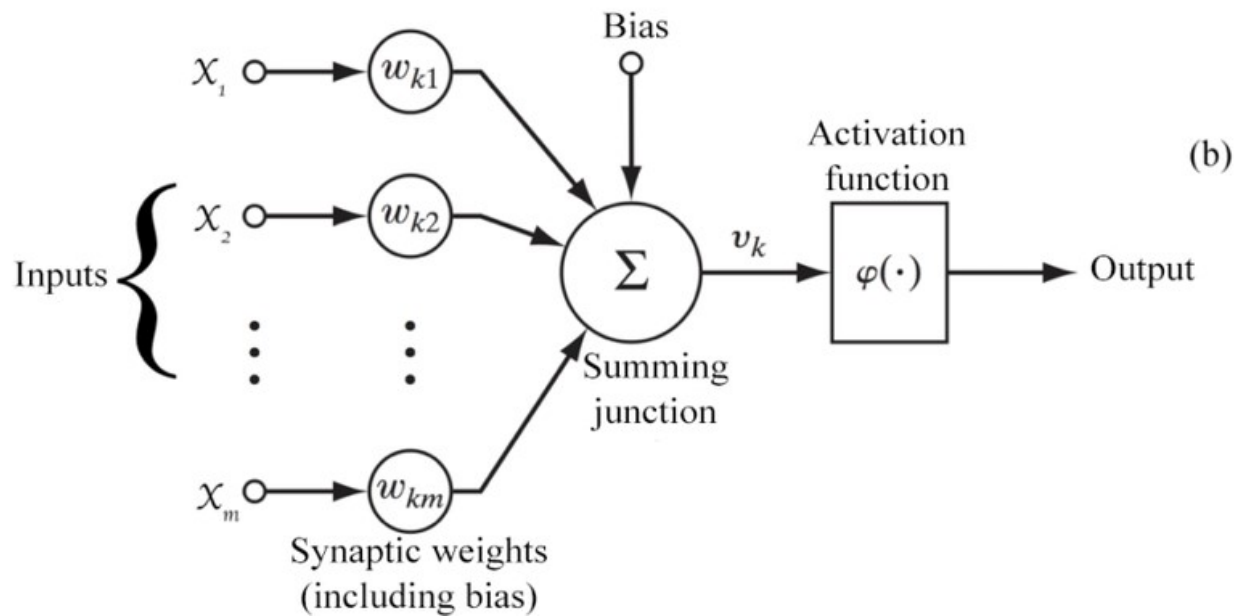
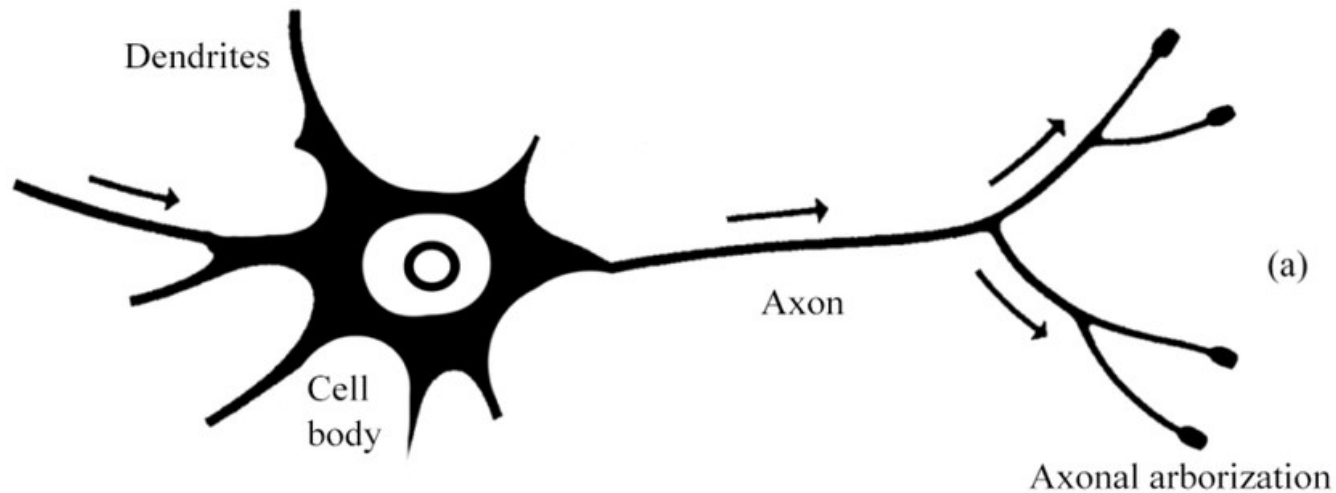
Types of Neural Networks

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Neural Networks



Artificial Neural Networks



Artificial Neural Networks

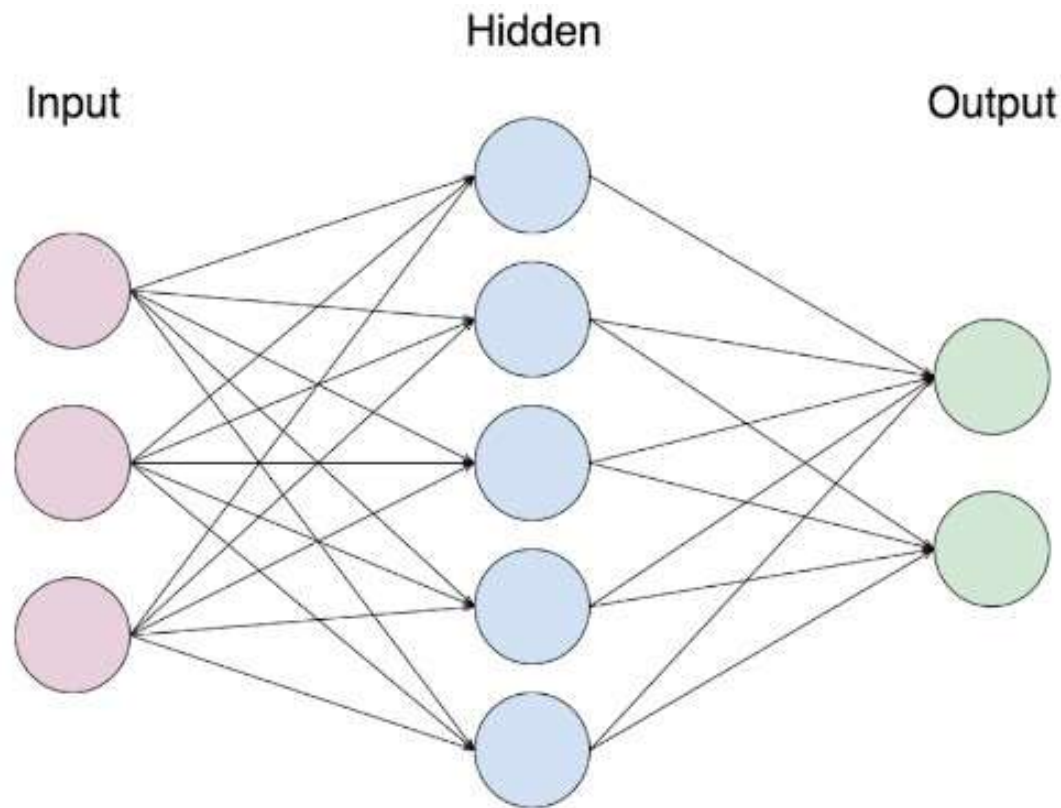
- An artificial neural network (ANN) is composed of artificial neurons or nodes.
- The connections of the biological neuron are modeled in artificial neural networks as weights between nodes.
- A positive weight reflects an excitatory connection, while negative values mean inhibitory connections.

Artificial Neural Networks

- An ANN consists of connected units or nodes called artificial neurons, which loosely model the neurons in the brain.
- These are connected by edges, which model the synapses in the brain.
- Each artificial neuron receives signals from connected neurons, then processes them and sends a signal to other connected neurons.
- The "signal" is a real number, and the output of each neuron is computed by some non-linear function of the sum of its inputs, called the activation function.
- The strength of the signal at each connection is determined by a weight, which adjusts during the learning process.

Architecture of Artificial Neural Network

- An Artificial Neural Network is made up of 3 components:



BNN vs. ANN

Parameter	Biological Neural Networks (BNNs)	Artificial Neural Networks (ANNs)
Basic Unit	Neuron: The fundamental cell responsible for processing and transmitting information in the brain.	Artificial Neuron (Node): Simplified mathematical models that simulate neuron functions.
Signal Transmission	Electrochemical signals via synapses: Neurons communicate using electrical impulses and chemical signals across synapses.	Numerical values via weighted connections: Nodes transmit information through numerical weights.

BNN vs. ANN

Learning Mechanism	Hebbian learning, synaptic plasticity: Learning involves changes in the strength of synapses based on activity patterns.	Backpropagation, gradient descent, etc.: Learning adjusts weights using algorithms to minimize error.
Processing Speed	Relatively slow (milliseconds to seconds per signal): Biological neurons have slower transmission speeds due to the nature of chemical and electrical processes.	Fast (microseconds to milliseconds per computation): Computational nodes process information rapidly, limited by hardware.

BNN vs. ANN

Energy Efficiency	Very efficient, low power consumption: The brain operates on about 20 watts, highly efficient for its complexity.	Less efficient, high computational power required: ANNs, especially large models, consume significant power.
Scalability	Naturally scalable and self-organizing: BNNs can grow and reorganize their connections as needed.	Requires significant resources for large-scale models: Scaling ANNs involves more hardware and computational power.

BNN vs. ANN

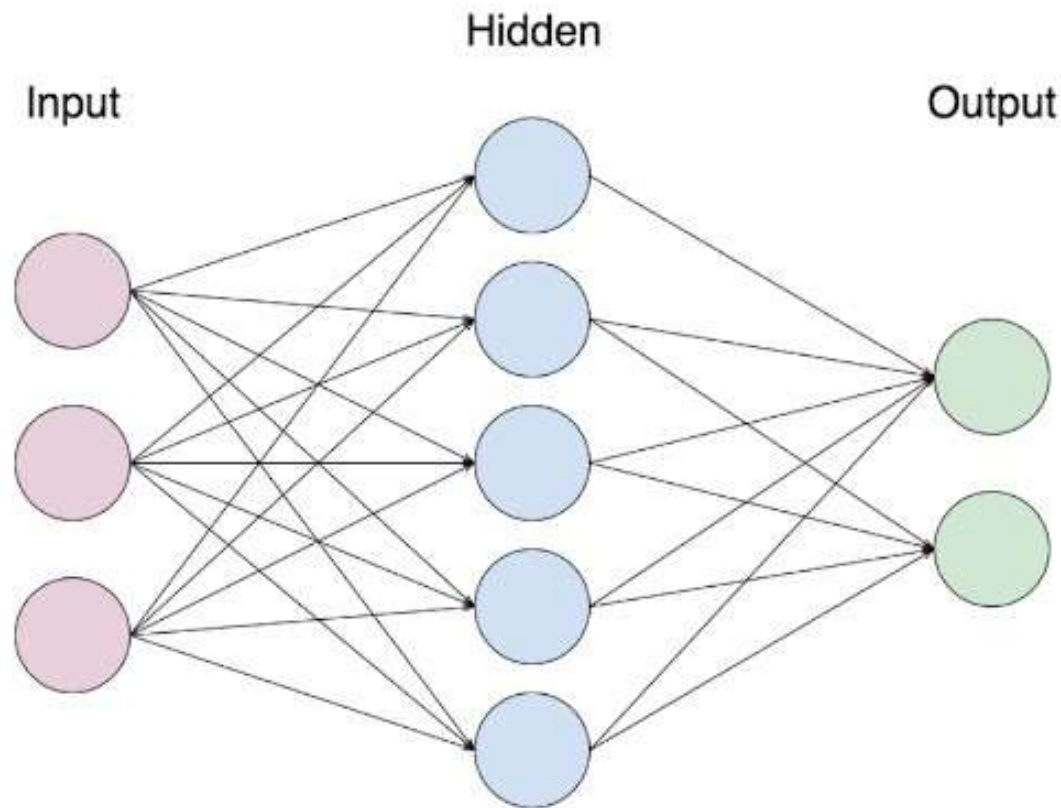
Plasticity	High, can rewire and adapt over time: The brain's structure and function can change significantly through neuroplasticity.	Low, fixed structure during inference after training: Once trained, the ANN's structure remains mostly static.
Information Encoding	Spike timing and frequency (temporal coding): Information is encoded in the timing and frequency of neuron spikes.	Continuous numerical values (usually in range $[-1, 1]$): Information is represented as numerical values in a continuous range.

BNN vs. ANN

Robustness	Highly robust, capable of coping with damage: The brain can often function effectively despite damage or loss of neurons.	Susceptible to performance degradation with changes: ANNs can be sensitive to changes in input data or architecture.
Parallel Processing	Naturally parallel, many neurons firing simultaneously: BNNs process information in a massively parallel manner.	Simulated parallelism using CPUs or GPUs: ANNs achieve parallelism through hardware that simulates concurrent operations.

Architecture of Simple Neural Networks

- An Artificial Neural Network is made up of 3 components:



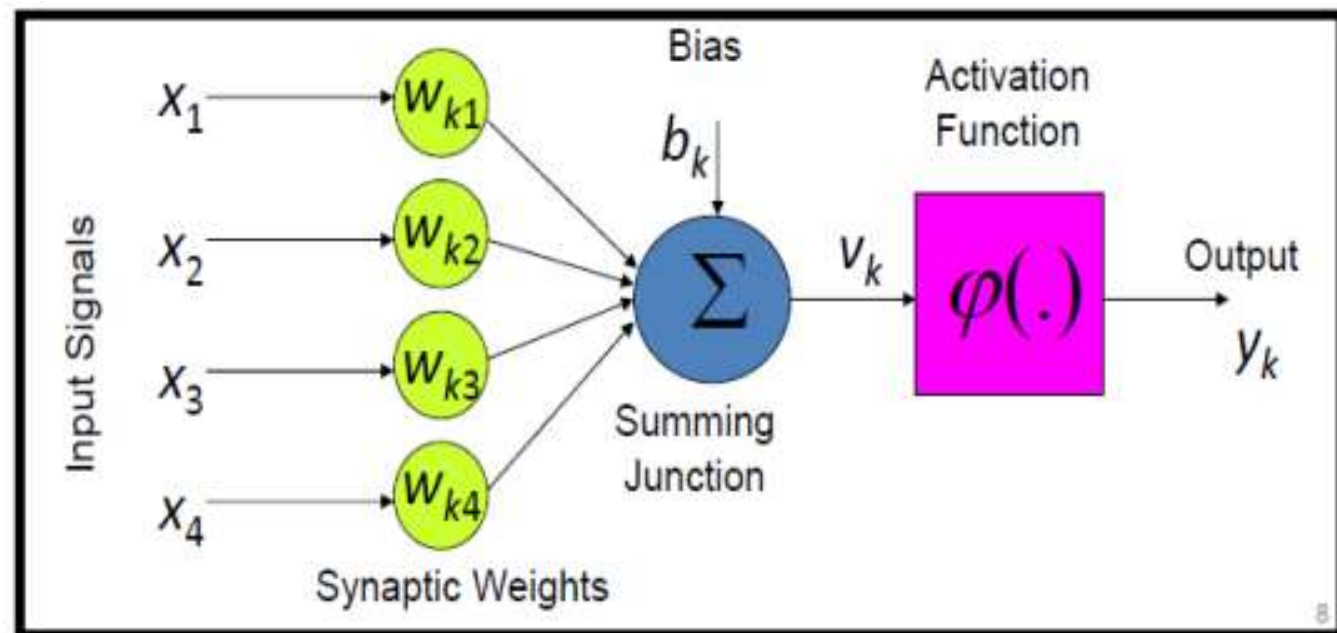
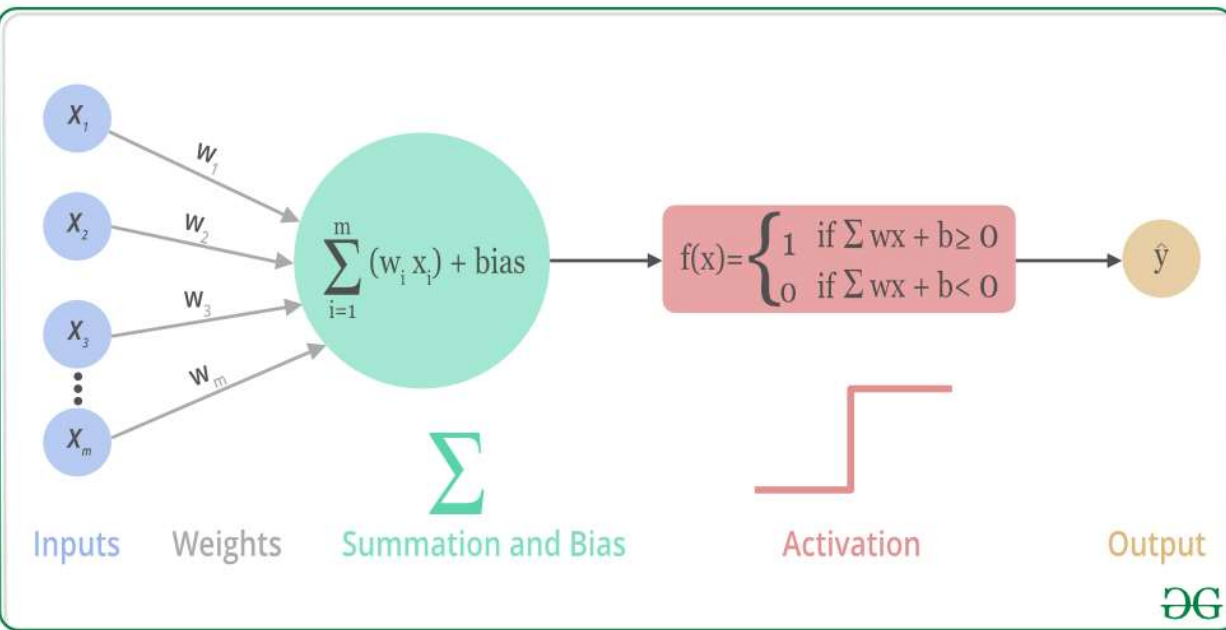
Architecture of Simple Neural Networks

- **Input Layer:** This is literally the layer that inputs information for the neural network to process. Each circle represents 1 feature (a piece of information).
- **Hidden Layer:** These layers do all the processing for neural networks. Each layer consists of nodes that mimic our brains' neurons. These nodes receive information from the previous layer's nodes, multiply it by **weight** and then add a **bias** to it.
- **Output Layer:** The output layer is the output/conclusions of the model derived from all the computations performed. There can be single or multiple nodes in the output layer.

Artificial Neural Networks

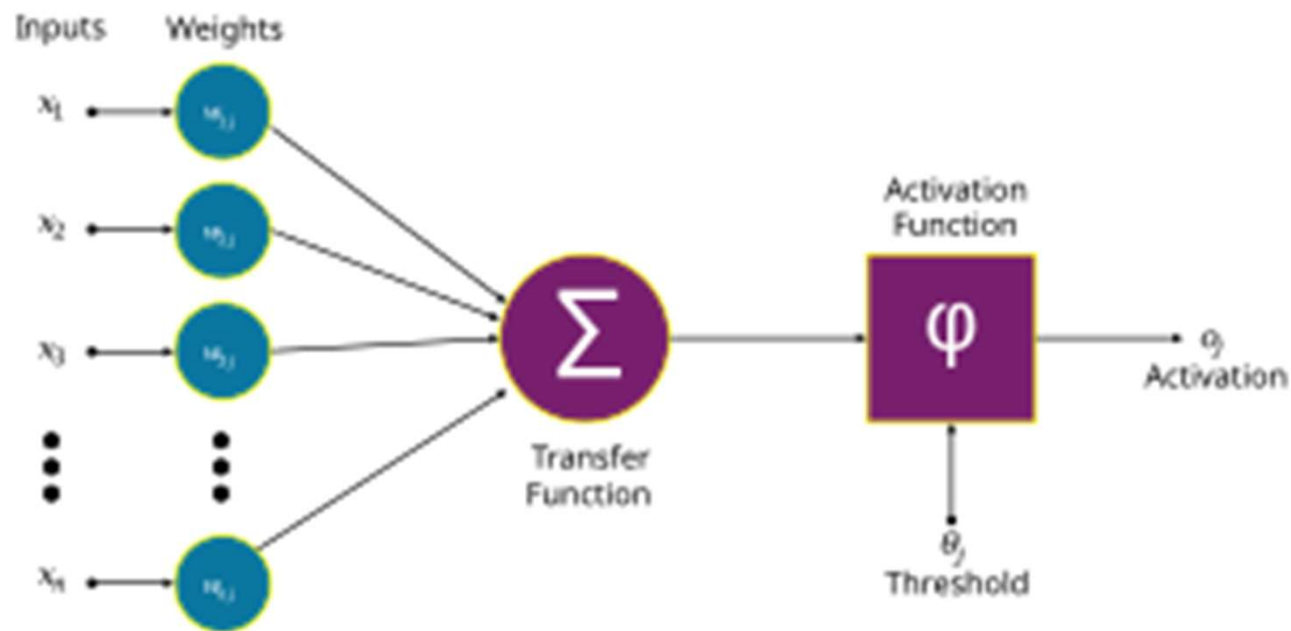
- A neural network is characterized by
 - Its pattern of connections between the neurons (called its architecture).
 - Its method of determining the weights on the connections (called its training, or learning, algorithm).
 - Its activation function.

Artificial Neural Networks



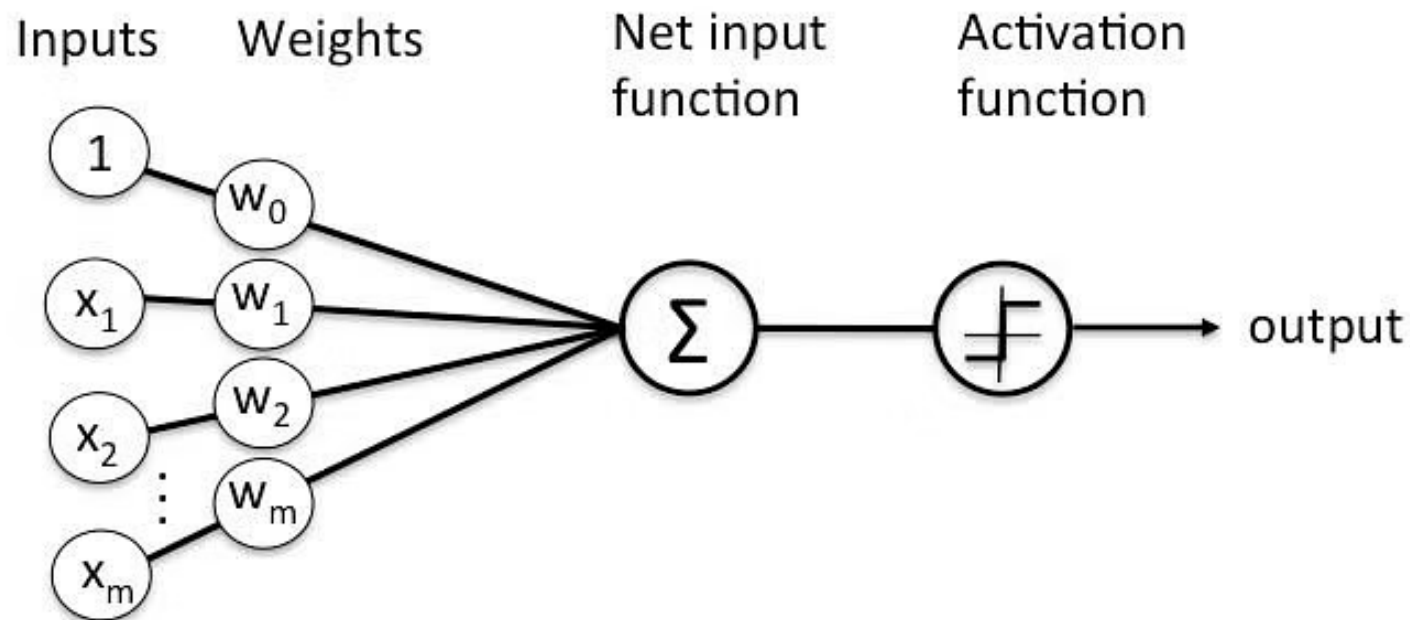
Artificial Neuron

- An artificial neuron is a mathematical function based on the model of biological neurons, where each neuron **takes inputs, weighs them separately, sums them up** and passes this sum through a function (linear or nonlinear) to produce output.



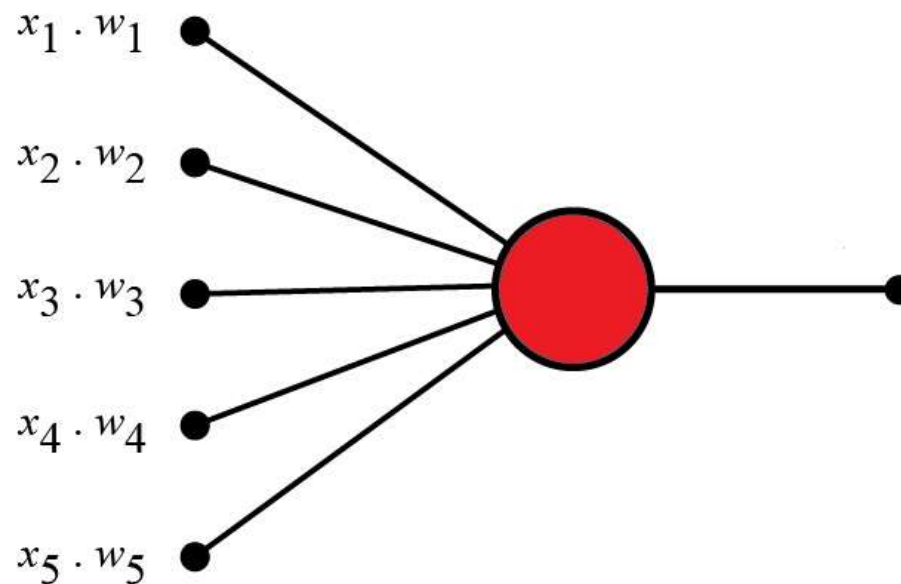
Perceptron

- A Perceptron is an Artificial Neuron. It is the simplest possible Neural Network.
- A Perceptron is an algorithm for supervised learning of binary classifiers. This algorithm enables neurons to learn and processes elements in the training set one at a time



Perceptron

- The original Perceptron was designed to take a number of binary inputs and produce one binary output (0 or 1).
- The idea was to use different weights to represent the importance of each input, and that the sum of the values should be greater than a threshold value before making a decision like yes or no (true or false) (0 or 1).



Perceptron

- **Input Layer:** The input layer consists of one or more input neurons, which receive input signals from the external world or from other layers of the neural network.
- **Weights:** Each input neuron is associated with a weight, which represents the strength of the connection between the input neuron and the output neuron.
- **Bias:** A bias term is added to the input layer to provide the perceptron with additional flexibility in modeling complex patterns in the input data.

Perceptron

- **Activation Function:** The activation function determines the output of the perceptron based on the weighted sum of the inputs and the bias term.
- **Output:** The output of the perceptron is a single binary value, either 0 or 1, which indicates the class or category to which the input data belongs.
- **Training Algorithm:** The perceptron is typically trained using a supervised learning algorithm such as the perceptron learning algorithm or backpropagation. During training, the weights and biases of the perceptron are adjusted to minimize the error between the predicted output and the true output for a given set of training examples.

Perceptron Example

- Imagine a perceptron (in your brain).
- The perceptron tries to decide if you should go to a concert.
- Is the artist good? Is the weather good?
- What weights should these facts have?

Perceptron Example

Criteria	Input	Weight
Artists is Good	$x_1 = 0 \text{ or } 1$	$w_1 = 0.7$
Weather is Good	$x_2 = 0 \text{ or } 1$	$w_1 = 0.6$
Friends will Come	$x_3 = 0 \text{ or } 1$	$w_1 = 0.5$
Food is Served	$x_4 = 0 \text{ or } 1$	$w_1 = 0.3$
Free Wifi is Available	$x_5 = 0 \text{ or } 1$	$w_1 = 0.4$

Perceptron Example

- Perceptron Algorithm:
 - Set a threshold value
 - Multiply all inputs with its weights
 - Sum all the results
 - Activate the output

Perceptron Example

Threshold value = 1.5

$$x_1 * w_1 = 1 * 0.7 = 0.7$$

$$x_2 * w_2 = 0 * 0.6 = 0$$

$$x_3 * w_3 = 1 * 0.5 = 0.5$$

$$x_4 * w_4 = 0 * 0.3 = 0$$

$$x_5 * w_5 = 1 * 0.4 = 0.4$$

$$0.7 + 0 + 0.5 + 0 + 0.4 = 1.6 \text{ (The Weighted Sum)}$$

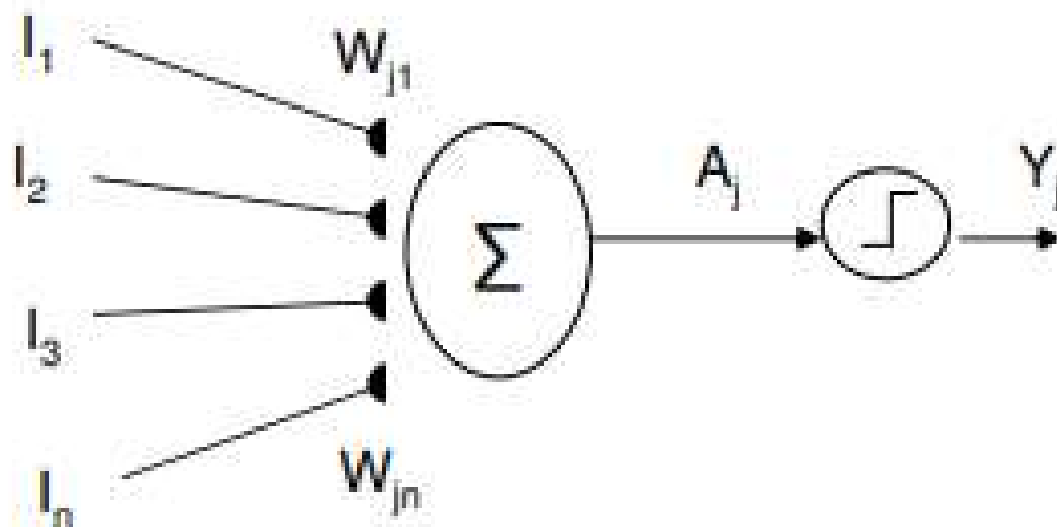
Return true if the sum > 1.5 ("Yes I will go to the Concert")

Single Layer Perceptron

- A Single Layer Perceptron (SLP) is the very basic form of Artificial Neural Network (ANN).
 - Taking all the input values and multiplying them by their weights.
 - Then, all of these multiplied values are added together to create the weighted sum.
 - The weighted sum is then applied to the activation function, producing the perceptron's output.
- It is important to note that the **weight of an input is indicative of the strength of a node**. Similarly, an input's **bias value gives the ability to shift the activation function curve up or down**. Bias is similar to Intercept in Linear Regression.

Single Layer Perceptron: Example

AND		
in_1	in_2	out
0	0	0
0	1	0
1	0	0
1	1	1



Single Layer Perceptron: Example

$$\Sigma = x_1 * w_1 + x_2 * w_2 = 0 * 0.9 + 0 * 0.9 = 0$$

Let's focus on the 2nd instance. $x_1 = 0$ and $x_2 = 1$.

$$\text{Sum unit: } \Sigma = x_1 * w_1 + x_2 * w_2 = 0 * 0.9 + 1 * 0.9 = 0.9$$

$$\varepsilon = \text{actual} - \text{prediction} = 0 - 1 = -1$$

$$w_1 = w_1 + \alpha * \varepsilon = 0.9 + 0.5 * (-1) = 0.9 - 0.5 = 0.4$$

$$w_2 = w_2 + \alpha * \varepsilon = 0.9 + 0.5 * (-1) = 0.9 - 0.5 = 0.4$$

α = is learning rate

Single Layer Perceptron: Example

Focus on the 3rd instance. $x_1 = 1$ and $x_2 = 0$.

Sum unit: $\Sigma = x_1 * w_1 + x_2 * w_2 = 1 * 0.4 + 0 * 0.4 = 0.4$

Mention the 4th instance. $x_1 = 1$ and $x_2 = 1$.

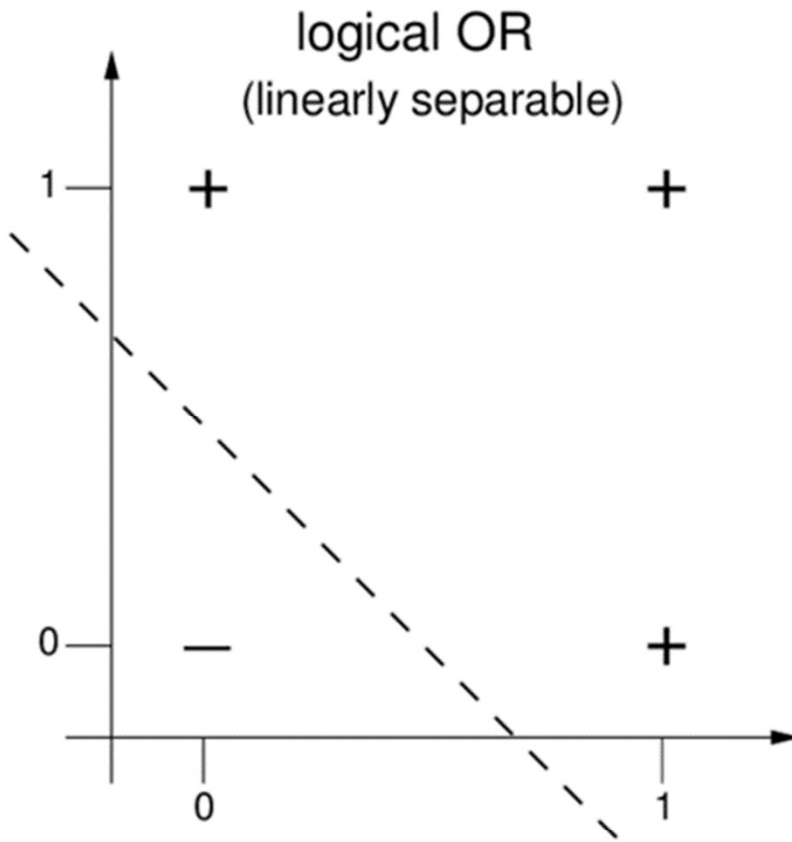
Sum unit: $\Sigma = x_1 * w_1 + x_2 * w_2 = 1 * 0.4 + 1 * 0.4 = 0.8$

Single Layer Perceptron: Exercise

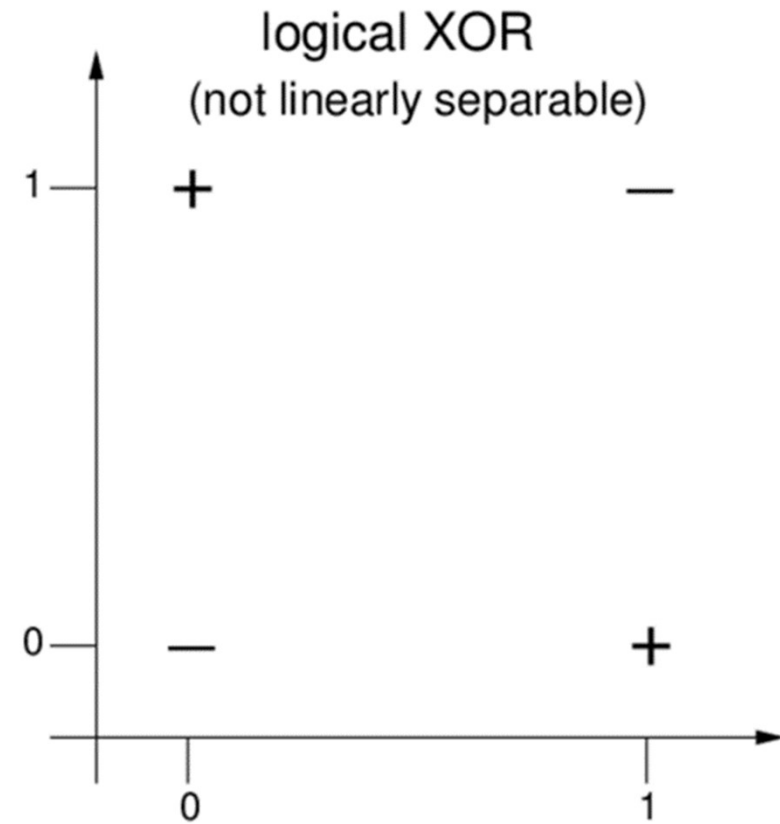
Implement the following function in a Single Layer Perceptron.

X_1	X_2	Y
0	0	0
0	1	1
1	0	1
1	1	1

Single Layer Perceptron: Exercise



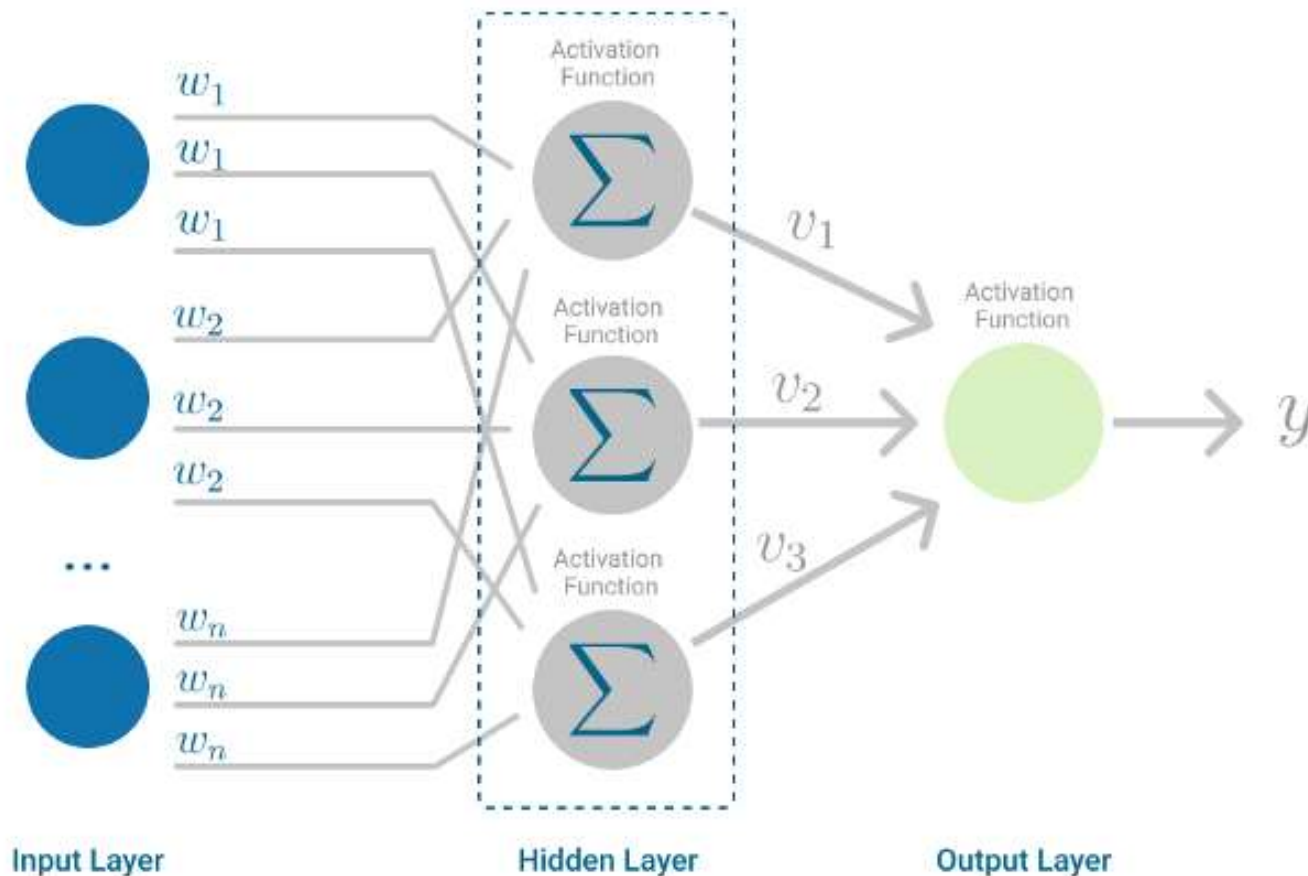
Input ₁	Input ₂	Output
1	1	1
1	0	1
0	1	1
0	0	0



Input ₁	Input ₂	Output
1	1	0
1	0	1
0	1	1
0	0	0

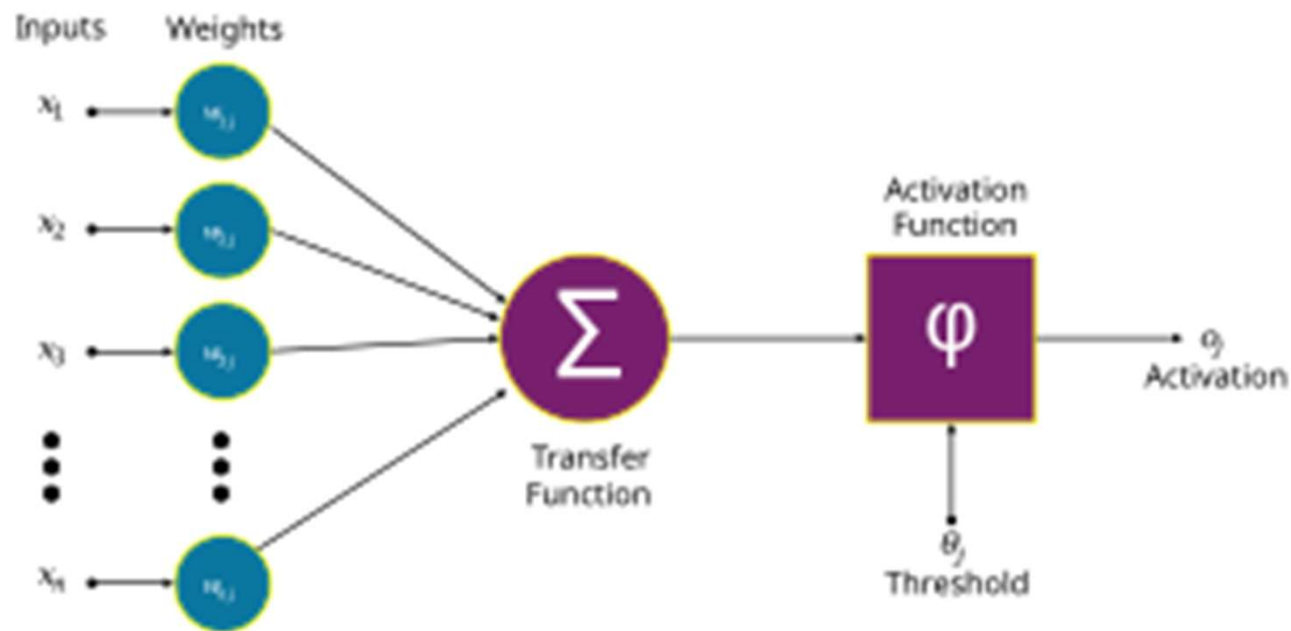
Multilayer Perceptron

- It is a neural network where the mapping between inputs and output is **non-linear**.
- A Multilayer Perceptron has input and output layers, and one or more hidden layers with many neurons stacked together.



Artificial Neuron

- An artificial neuron is a mathematical function based on a model of biological neurons, where each neuron **takes inputs, weighs them separately, sums them up** and passes this sum through a function (linear or **nonlinear**) to produce output.



Artificial Neuron

- A neuron is a mathematical function modeled on the working of biological neurons.
- It is an elementary unit in an artificial neural network.
- One or more inputs are separately weighted.
- Inputs are summed and passed through a **nonlinear function** to produce output.
- Every neuron holds an internal state called activation signal.
- Each connection link carries information about the input signal.
- Every neuron is connected to another neuron via connection link.

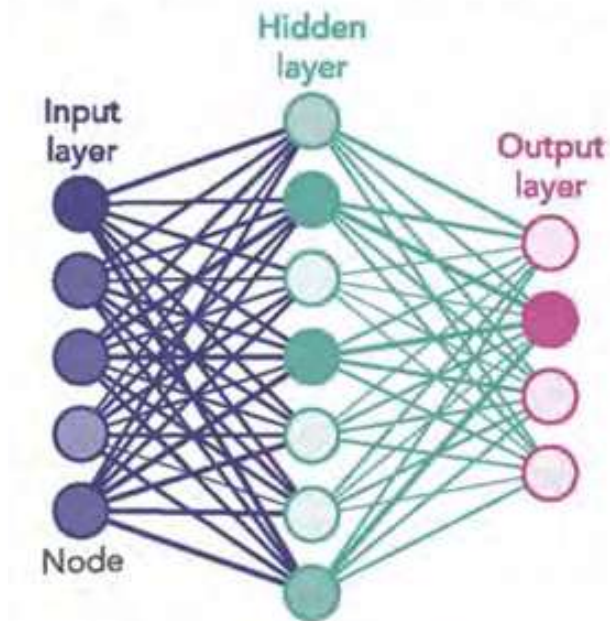
Perceptron vs. Neuron

- Neuron employs non-linear activation function and perceptron employs only a threshold (linear) activation function.
- The output of a neuron is not necessarily a binary number and the output of a perceptron is always a binary number.
- A perceptron can be thought of as a standalone model while the artificial neuron is the smallest computational unit of a neural network.

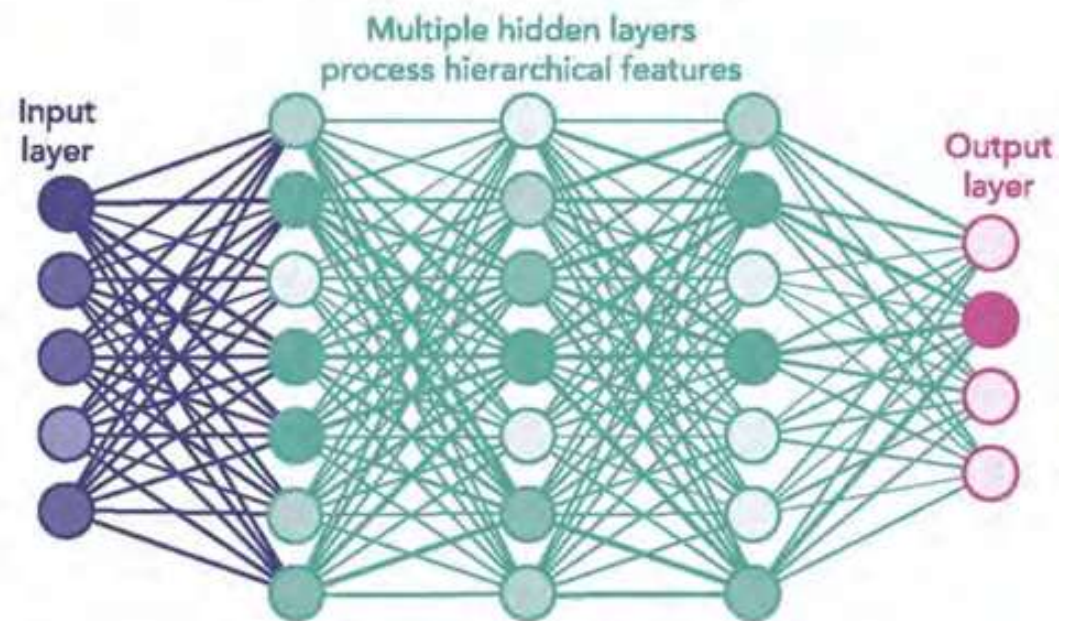
Types of Artificial Neural Network

- Shallow neural networks usually have only one hidden layer.
- Deep neural networks have multiple hidden layers.

SHALLOW NEURAL NETWORK

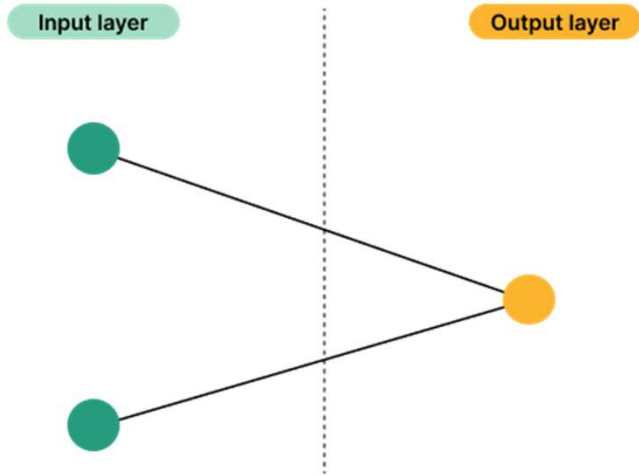


DEEP NEURAL NETWORK

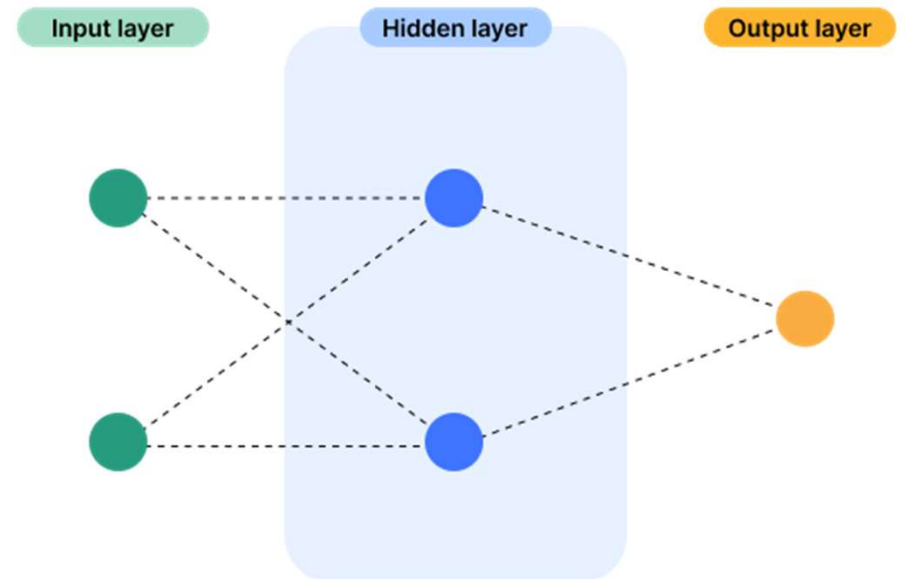


Types of Artificial Neural Network

Perceptron

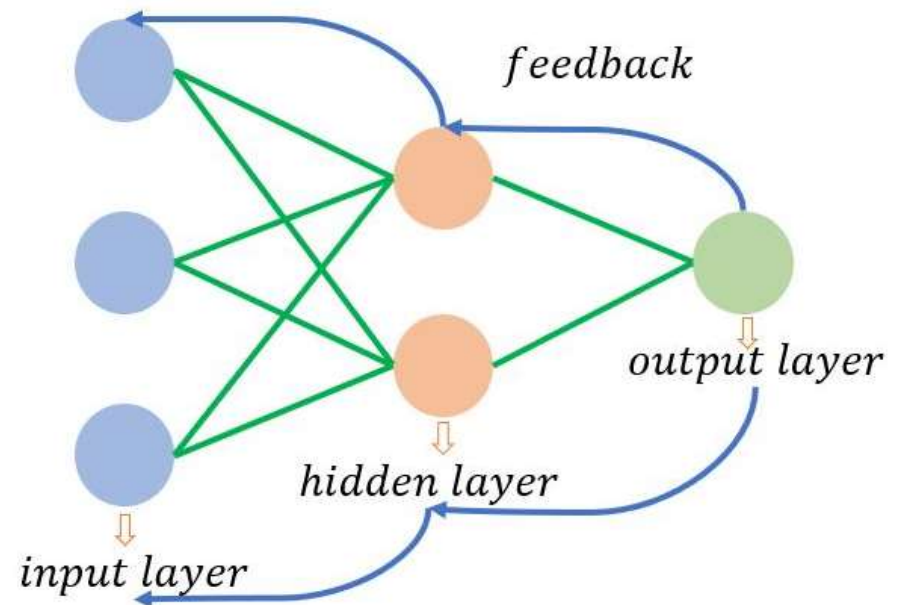
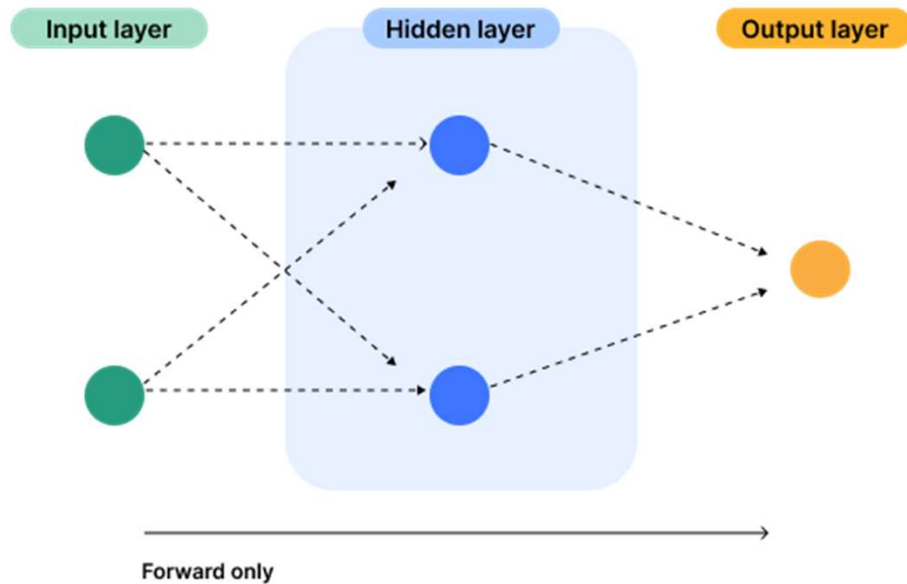


Multilayer perceptron



Types of Artificial Neural Network

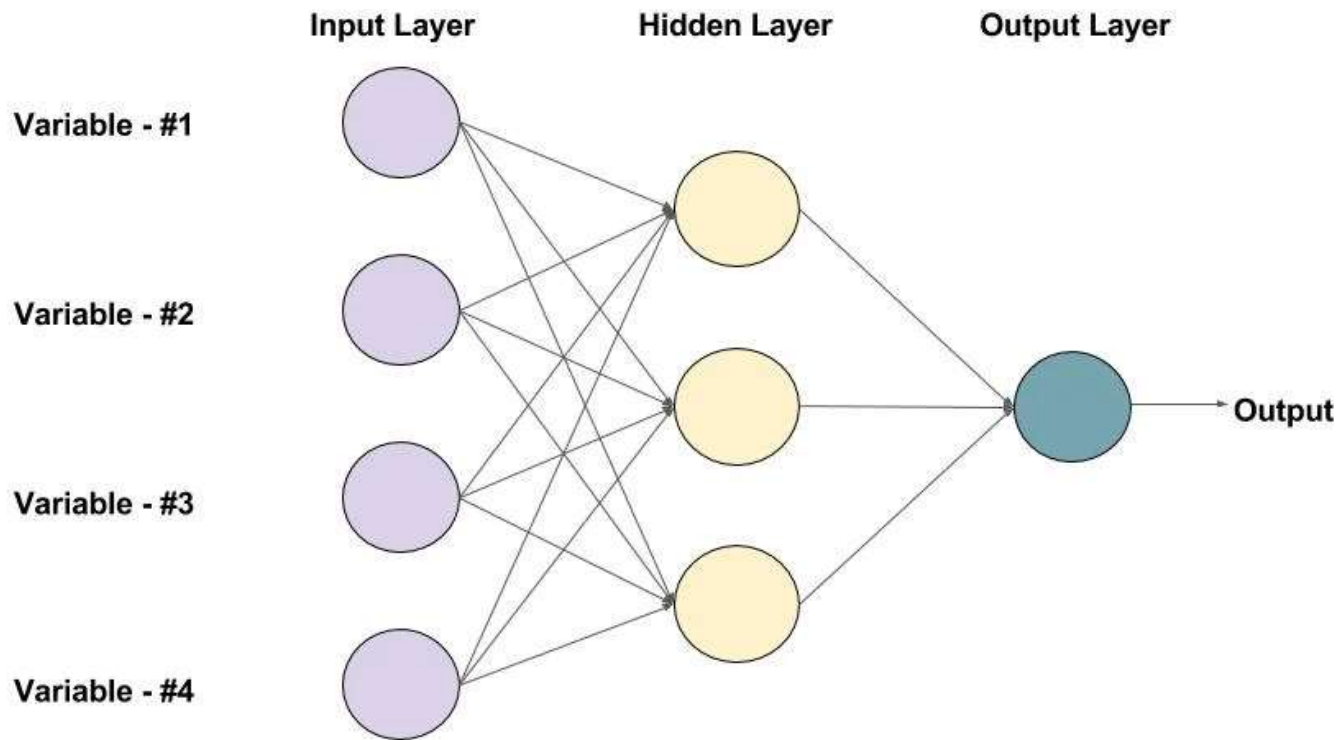
Feed-forward



Feedforward Neural Networks (FNN)

- In these types of neural networks information flows in only one direction, i.e., from input layer to output layer.
- When the weights are once decided, they are not usually changed.
- One either explicitly decides weights or uses functions like Radial Basis Function to decide weights.
- The nodes here do their job without being aware whether results produced are accurate or not(i.e., they don't re-adjust according to result produced).
- There is no communication back from the layers ahead.

Feedforward Neural Networks

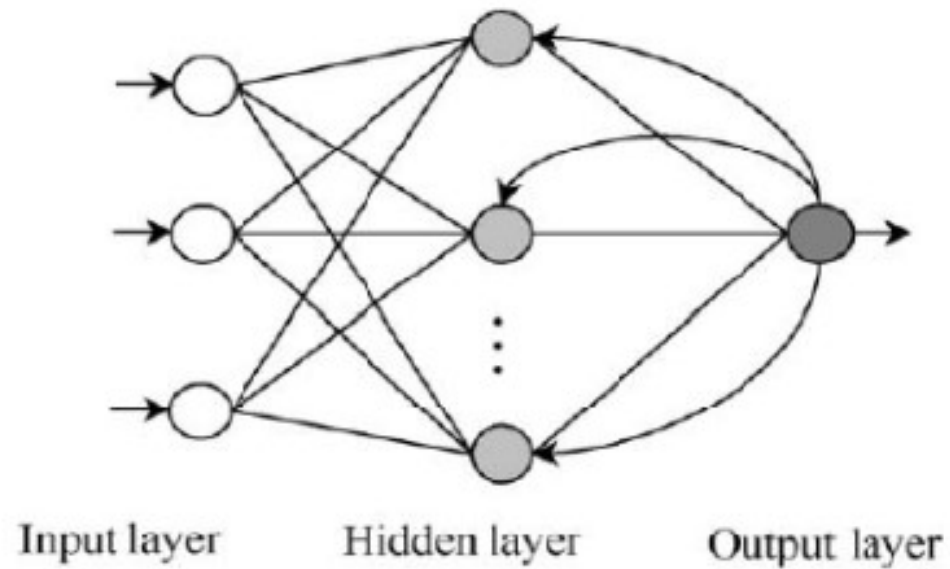


An example of a Feed-forward Neural Network with one hidden layer (with 3 neurons)

Feedback Neural Networks

- Signals can travel in both the directions in Feedback neural networks.
- Feedback neural networks are very powerful and can get very complicated.
- Feedback neural networks are dynamic.
- The ‘state’ in such network keep changing until they reach an equilibrium point.
- They remain at the equilibrium point until the input changes and a new equilibrium needs to be found.

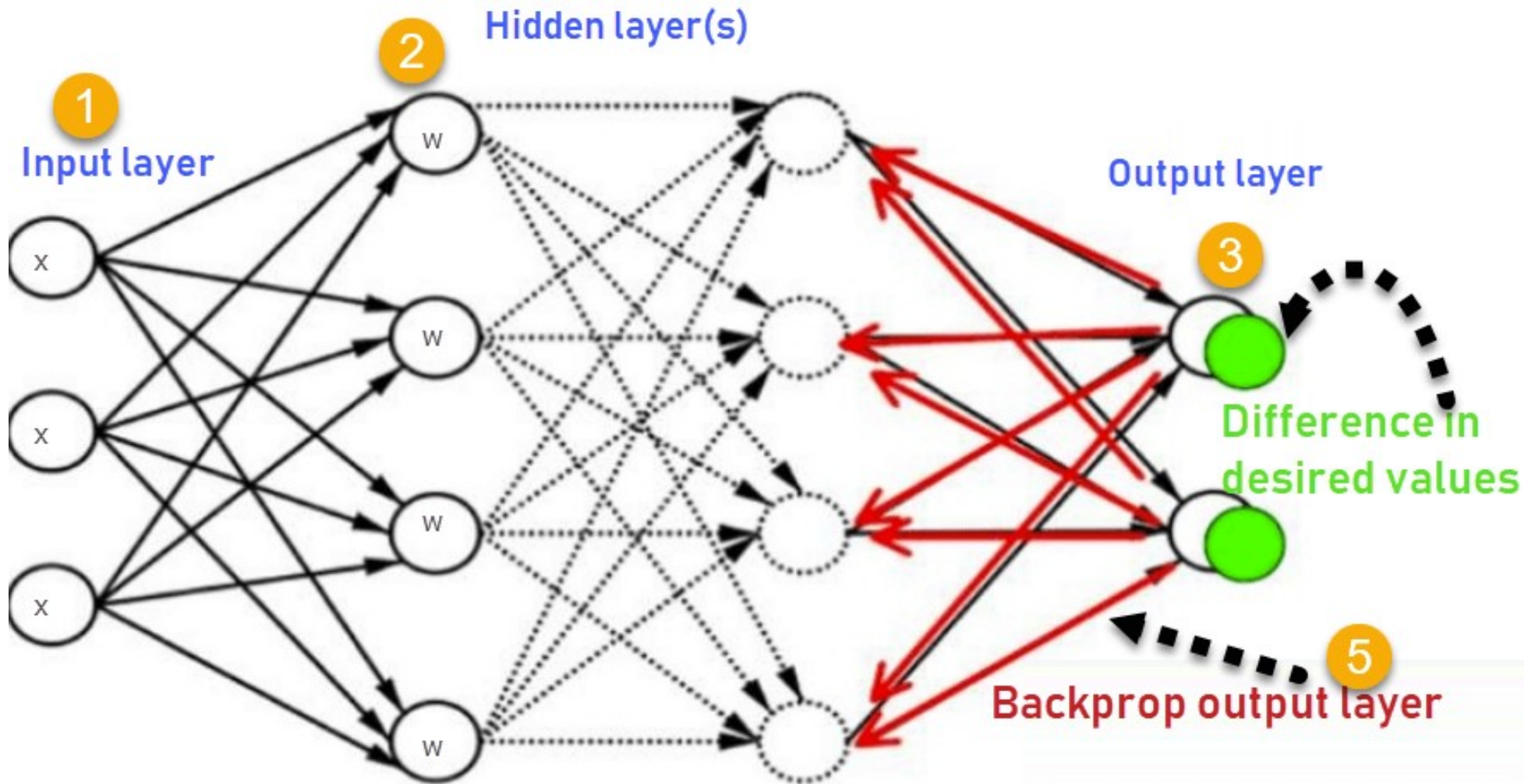
Feedback Neural Networks



How Neural Networks learn?

- The training samples are passed through the network and the output obtained from the network is compared with the actual output and **error is calculated**.
- **Backpropagation** in neural network is a short form for “backward propagation of errors.” It is a standard method of training artificial neural networks. This method helps calculate the gradient of a loss function with respect to all the weights in the network.
- Backpropagation is the essence of neural network training. It is the method of fine-tuning the weights of a neural network based on the error rate obtained in the previous epoch (i.e., iteration). Proper tuning of the weights allows us to reduce error rates and make the model reliable by increasing its generalization.

How Neural Networks learn?



Backpropagation Pros and Cons

- Backpropagation is fast, simple and easy to program.
- It is a flexible method as it does not require prior knowledge about the network.
- It does not need any special mention of the features of the function to be learned.
- The actual performance of backpropagation on a specific problem is dependent on the input data.
- Back propagation algorithm in data mining can be quite sensitive to noisy data.

Learning Rate in Neural Networks

- The learning rate is a hyperparameter that controls how much to change the model in response to the estimated error each time the model weights are updated.
- Choosing the learning rate is challenging as a value too **small may result in a long training process** that could get stuck, whereas a **value too large may result in learning a sub-optimal set of weights** too fast or an unstable training process

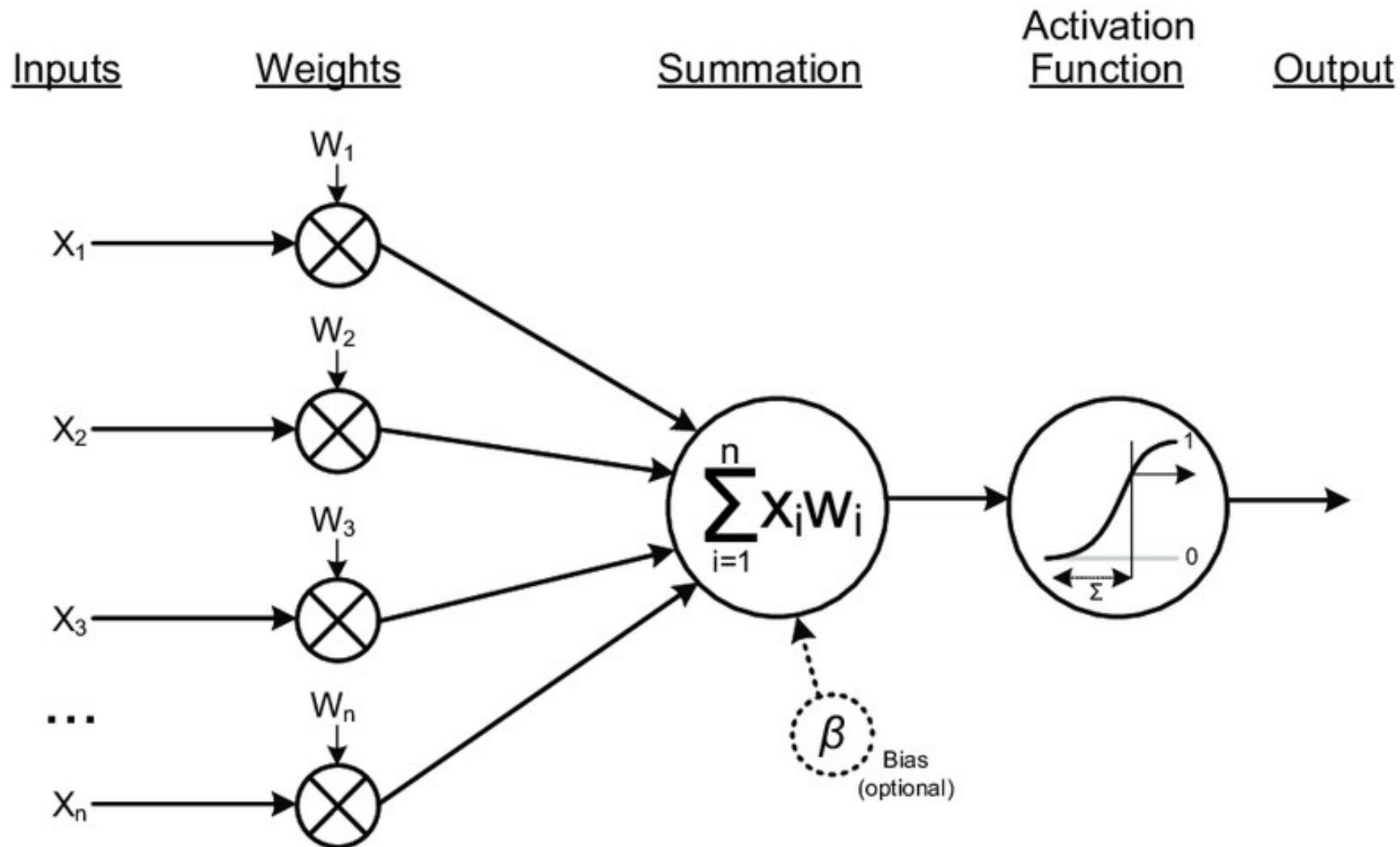
Learning Rate in Neural Networks

- Generally, stochastic gradient descent is an optimization algorithm that **estimates the error gradient** for the current state of the model using examples from the training dataset, then updates the weights of the model using the back-propagation.
- The **amount that the weights are updated** during training is referred to as the step size or the “learning rate.”
- The learning rate is a **configurable hyperparameter** used in the training of neural networks that has a small positive value, often in the range between **0.0 and 1.0**.

Activation Function in Neural Networks

- An activation function in a neural network defines how the weighted sum of the input is transformed into an output from a node or nodes in a layer of the network.
- Activation function decides, whether a neuron should be activated or not by calculating weighted sum and further adding bias with it.
- The purpose of the activation function is to introduce non-linearity into the output of a neuron.

Activation Function in Neural Networks



Summary

- Neural Networks